The degree-six polynomial in Helton–Klep–McCullough–Volčič is minimal

An exact computer-assisted full solution of the degree-five-versus-six question

Autonomous proof search, run `fa_banach_001`, lane 2

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Status. Claimed full solution, likely valid, pending expert review. The finite algebraic classification in the proof is certified exactly over $\mathbb{Q}(\sqrt{2})$ by the included verifier; no floating-point computation is used.

1 Source question

The source is J. W. Helton, I. Klep, S. McCullough and J. Volčič, *Noncommutative polynomials describing convex sets*, arXiv:1808.06669, Section 5.2, physical PDF page 24 [1].

$$\{(X, X^*) \colon f(X, X^*) \geq 0\} \neq \mathcal{D}_L$$

in this case.

5.2. **Example of degree 5 or 6.** Let

$$\begin{aligned} f_1 &= 1 - (x + x^*) - 2(x + x^*)^2 - 2x^*x + (x + x^*)^3 + 2(x + x^*)^2 x^* x, \\ s &= 1 - (x + x^*)^2 \end{aligned}$$

and

$$L = \begin{pmatrix} 1 - \frac{1}{2}(x + x^*) & -\sqrt{2}(x + x^*) & \frac{1}{2}(x + x^*) & x^* \\ -\sqrt{2}(x + x^*) & 1 & 0 & 0 \\ \frac{1}{2}(x + x^*) & 0 & 1 - \frac{1}{2}(x + x^*) & -x^* \\ x & 0 & -x & 1 \end{pmatrix}.$$

As in the previous example the only item of Lemma 5.1 that is not simple to verify is (5). Observe that the upper 2×2 block of L depends only on the hermitian variable $h = x + x^*$. The same holds for $s = 1 - h^2$. Hence it suffices to see that $s > 0$ on $\mathcal{D}_L(1)$, which is true since

$$\det \begin{pmatrix} 1 - \frac{\rho}{2} & -\sqrt{2}\rho \\ -\sqrt{2}\rho & 1 \end{pmatrix} \geqslant 0 \quad \Longrightarrow \quad 1 - \rho^2 > 0$$

for $\rho \in \mathbb{R}$. If $f = f_1 s$, then $\mathcal{D}_f$ is a free spectrahedron domain whose minimal degree defining polynomial has degree at least 5. Note that $\deg f = 6$, but we do not know whether f is a minimal degree defining polynomial.

Of course, by taking a Schur complement of L we obtain a quadratic 2×2 noncommutative polynomial q with $\mathcal{D}_q = \mathcal{D}_L$:

Put $y = x^*$ and $H = x + y$. The source defines

$$\begin{aligned} f_1 &= 1 - H - 2H^2 - 2yx + H^3 + 2H^2yx, \\ s &= 1 - H^2, \end{aligned}$$

and

$$L = \begin{pmatrix} 1 - \frac{1}{2}H & -\sqrt{2}H & \frac{1}{2}H & y \\ -\sqrt{2}H & 1 & 0 & 0 \\ \frac{1}{2}H & 0 & 1 - \frac{1}{2}H & -y \\ x & 0 & -x & 1 \end{pmatrix}.$$

It proves that $f = f_1 s$ is Hermitian, $\mathcal{D}_f = \mathcal{D}_L$, $\deg f = 6$, and every scalar Hermitian defining polynomial for this domain has degree at least five. The exact source sentence is:

> "Note that $\deg f = 6$, but we do not know whether f is a minimal degree defining polynomial."

Thus the missing issue is the existence or nonexistence of a degree-five scalar Hermitian polynomial defining $\mathcal{D}_L$.

2 Definitions and input results

We work in $\mathbb{C}\langle x, y\rangle$ with involution $x^* = y$, $y^* = x$. For a scalar polynomial q with $q(0) = 1$, $\mathcal{K}_q$ is the closure of the connected component of the origin in the set of matrix tuples where q is invertible. When $q = q^*$, the same set is denoted $\mathcal{D}_q$. The free singularity locus is $\mathcal{Z}_q$.

We use the following results exactly as they are used in the source paper.

1. Lemma 5.1 of [1], together with the Polynomial Singularitätstellensatz, says that any Hermitian q with $\mathcal{D}_q = \mathcal{D}_L$ has degree at least five; moreover, a degree-five such q has an atomic factor of degree four in the stable-association class of f_1 [2, Theorem 4.3].

2. Proposition 2.7 and Remark 2.6 of [1] give the minimal Fornasini–Marchesini realization, the determinant identity, the inversion formula, and the joint-nilpotence criterion for a realization to be a polynomial.

3. The proof of Proposition 3.3 of [1] applies verbatim to an arbitrary normalized atom p with $\mathcal{Z}_p = \mathcal{Z}_L$: a minimal realization of p^{-1} can be written over the displayed pencil L. Indeed, its irreducible realization pencil has the same free locus as L and hence is similar to L by [3, Theorem 3.11].

3 Proof intuition

The source lower bound leaves only one possible way a degree-five defining polynomial could exist: it would have to be a degree-four atom with the same free singularity locus as f_1, multiplied by an affine-linear factor. Stable association means that the degree-four atom need not literally be f_1, so checking only $f_1 t$ is insufficient.

Minimal realization theory makes all stable associates finite in this example. A degree-four atom p with $\mathcal{Z}_p = \mathcal{Z}_L$ gives a rank-one matrix $M = v\alpha$ satisfying the scalar adjugate identity

$$\operatorname{tr}(M \operatorname{adj} L(z, \bar{z})) = 1.$$

Coefficient comparison leaves a seven-dimensional affine space, and the rank-one minors cut it down to exactly four points. These yield four explicit degree-four atoms (one is f_1). For each atom, six coefficients of $pt - (pt)^*$ form a nonsingular linear system in the six real and imaginary parts of an affine t. Hence $t = 0$, contradicting a degree-five factorization.

4 Main theorem

Theorem 1. *Let f_1, s, L be the polynomials and pencil above and set $f = f_1 s$. If $q \in \mathbb{C}\langle x, y\rangle$ is Hermitian, $q(0) = 1$, and $\mathcal{D}_q = \mathcal{D}_L$, then $\deg q \geq 6$. Consequently f is a minimal-degree scalar Hermitian defining polynomial for $\mathcal{D}_L$, and the minimum degree is exactly six.*

Proof. The source paper proves $f = f^*$, $\mathcal{D}_f = \mathcal{D}_L$, and $\deg f = 6$, as well as the lower bound $\deg q \geq 5$ [1, Lemma 5.1 and Section 5.2]. It remains to exclude degree five.

1. Reduction to a linear symmetrizer. Suppose $q = q^*$, $\mathcal{D}_q = \mathcal{D}_L$, and $\deg q = 5$. Normalize every factor to have constant term one. The argument of Lemma 5.1 in [1], using [2, Theorem 4.3(3)], gives an atomic factor p of degree four with

$$\mathcal{Z}_p = \mathcal{Z}_{f_1} = \mathcal{Z}_L.$$

Degree is additive in the free algebra, so all remaining atomic factors have total degree one. If p occurs on the left, $q = pt$ for a nonzero affine-linear t. If it occurs on the right, take adjoints and use $q = q^*$ to obtain the same form $q = p^* t^*$; here p^* has the same free singularity locus. Thus it suffices to prove:

$$\text{no normalized degree-four atom } p \text{ with } \mathcal{Z}_p = \mathcal{Z}_L \text{ has } pt = (pt)^* \text{ for nonzero affine } t. \tag{1}$$

2. Realization of every possible p. Write

$$L = I - A_x x - A_y y, \quad A_x = \begin{pmatrix} \frac{1}{2} & \sqrt{2} & -\frac{1}{2} & 0 \\ \sqrt{2} & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ -1 & 0 & 1 & 0 \end{pmatrix}, \quad A_y = A_x^{\mathsf{T}}.$$

By the realization results recalled above, every p in (1) has a minimal realization

$$p^{-1} = 1 + \alpha L^{-1}(b_x x + b_y y),$$

where α is a row vector. The inversion formula says that the realization coefficients for p are

$$N_x = A_x - b_x\alpha, \qquad N_y = A_y - b_y\alpha.$$

Since p is a polynomial, N_x, N_y are jointly nilpotent [1, Remark 2.6(5)–(6)]. Choose a nonzero common kernel vector v. If $\alpha v = 0$, then $A_x v = A_y v = 0$, contradicting minimality by Remark 2.6(1) of [1]. Scale v so that $\alpha v = 1$. Then

$$b_x = A_x v, \qquad b_y = A_y v, \qquad p^{-1} = \alpha L^{-1} v.$$

Proposition 2.7(1) of [1], evaluated at commuting scalar variables $z, \bar{z}$, gives $p(z, \bar{z}) = \det L(z, \bar{z})$. Hence

$$\alpha \operatorname{adj} L(z, \bar{z}) v = 1. \tag{2}$$

Set $M = v\alpha$. Then M has rank one and (2) is

$$\operatorname{tr}(M \operatorname{adj} L(z, \bar{z})) = 1. \tag{3}$$

3. Exact rank-one classification. Equating the coefficients in (3) gives every solution in the form

$$M(a,b,c,d,e,r,g) = \begin{pmatrix} 1-b+r/\sqrt{2} & e/\sqrt{2}+g/(2\sqrt{2})-a & g-b-c-1 & e \\ a & b & -e/\sqrt{2}-g/(2\sqrt{2})-d & r \\ c & d & -r/\sqrt{2}-g & g/2 \\ e & r & g/2 & g \end{pmatrix}.$$

The lexicographic Gröbner basis of its 2×2 minors contains

$$e, r, g, d^2, cd, c^2(c+1), c(b+c+1), d(b+1),$$
$$ac+2d, \quad b^2+b-2c^2-2c, \quad ab+a-4d, \quad a^2+2b-2c^2-2c.$$

The complete basis is printed by the verifier. These equations force $d=e=r=g=0$. If $c=-1$, they give $(a,b)=(0,0)$; if $c=0$, they give $(a,b)=(0,0)$ or $(\pm\sqrt{2},-1)$. Substitution verifies that these four points satisfy all minors. Therefore the only rank-one matrices are

$$M_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$M_3 = \begin{pmatrix} 2 & \sqrt{2} & 0 & 0 \\ -\sqrt{2} & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad M_4 = \begin{pmatrix} 2 & -\sqrt{2} & 0 & 0 \\ \sqrt{2} & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Choose factorizations $M_i = v_i \alpha_i$ as follows:

i	v_i	α_i
1	$(1,0,-1,0)^\mathsf{T}$	$(1,0,0,0)$
2	$(1,0,0,0)^\mathsf{T}$	$(1,0,-1,0)$
3	$(\sqrt{2},-1,0,0)^\mathsf{T}$	$(\sqrt{2},1,0,0)$
4	$(\sqrt{2},1,0,0)^\mathsf{T}$	$(\sqrt{2},-1,0,0).$

For $j \in \{x,y\}$ put

$$N_{i,j} = A_j(I-M_i), \qquad b_{i,j} = A_j v_i, \qquad N_i = N_{i,x}x + N_{i,y}y.$$

Direct exact multiplication gives $N_{i,w}=0$ for every word w of length four. Thus the corresponding polynomial is

$$p_i = 1 - \alpha_i(I + N_i + N_i^2 + N_i^3)(b_{i,x}x + b_{i,y}y). \tag{4}$$

Each p_i has degree four, and $p_2 = f_1$. Equations (3)–(4) show that every possible p is one of these four.

4. No affine symmetrizer. Let

$$t = u_0 + u_x x + u_y y, \qquad u_0 = u_{0r} + iu_{0i}, \quad u_x = u_{xr} + iu_{xi}, \quad u_y = u_{yr} + iu_{yi}.$$

For each i, equate coefficients in $p_i t - (p_i t)^* = 0$. The following six coefficients already have full rank. A word such as $xxxyx$ denotes the corresponding word in the free letters x, y.

i	Six coefficient equations
1	$-2u_{yr}=0$, $2u_{yi}=0$, $2u_{xr}=0$, $2u_{xi}=0$, $-2u_{0r}+u_{xr}-u_{yr}=0$, $2u_{0i}+u_{xi}+u_{yi}=0$
2	$2u_{xr}=0$, $2u_{xi}=0$, $2u_{yr}=0$, $2u_{yi}=0$, $2u_{0r}+u_{xr}-u_{yr}=0$, $2u_{0i}+u_{xi}+u_{yi}=0$
3	$-2u_{yr}=0$, $2u_{yi}=0$, $2u_{xr}=0$, $2u_{xi}=0$, $4u_{0i}-u_{xi}-u_{yi}=0$, $-4u_{0r}-2u_{xr}+4u_{yr}=0$
4	$-2u_{yr}=0$, $2u_{yi}=0$, $2u_{xr}=0$, $2u_{xi}=0$, $4u_{0i}+3u_{xi}+3u_{yi}=0$, $4u_{0r}-2u_{xr}+4u_{yr}=0$

For transparency, the selected words and real/imaginary parts are printed by the verifier. The determinants of these four 6×6 systems are respectively

$$64, \qquad 64, \qquad -256, \qquad 256.$$

Therefore $u_0=u_x=u_y=0$ for every i. This proves (1), contradicting the nonzero affine factor in a hypothetical degree-five q.

Hence every Hermitian defining polynomial has degree at least six. Since the source polynomial $f=f_1s$ has degree six and defines $\mathcal{D}_L$, it is minimal. □

5 Exact verification

The included script `code/verifier.py` performs the following exact checks over $\mathbb{Q}(\sqrt{2})$:

1. f_1s is Hermitian and has degree six;
2. the affine parametrization solves (3);
3. all 36 rank-one minors have exactly the four solutions above, with the displayed Gröbner basis;
4. all words of length four in every pair $N_{i,x}, N_{i,y}$ vanish;
5. every p_i has degree four and $p_2=f_1$;
6. the coefficient matrix for $p_it-(p_it)^*$ has rank six, with the four nonzero determinants displayed above.

Run it from the repository root with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/full/\
  1808.06669_degree_six_minimal_defining_polynomial/code/verifier.py
```

The recorded output ends with

```
PASS: exactly four realization candidates; no affine symmetrizer exists.
```

The computation is part of the proof only as an exact finite polynomial calculation; it is not numerical evidence.

6 Limitations, novelty check, and review recommendation

This result concerns the single explicit scalar example in Section 5.2 of [1]; it does not classify minimal defining degrees for arbitrary free spectrahedra. The proof relies on the cited realization and factorization theorems and on an exact finite Gröbner-basis/coefficient-rank certificate.

On 11 August 2026, a bounded novelty search checked the run's registry, solution, attempt, and proof-gap indexes and searched arXiv by the exact source sentence, the phrases "minimal degree defining polynomial" and "free spectrahedron," the title and arXiv id 1808.06669, and combinations of the authors' names with the degree-five-versus-six terminology. Only the source paper and background factorization papers were found; no later paper explicitly resolving this example was located. This is evidence of novelty, not a guarantee of priority.

Human review should focus on two points: (i) the passage from a degree-five Hermitian defining polynomial to a degree-four atomic factor realized over L, and (ii) the completeness of the four-point rank-one classification. Both are isolated explicitly above, and the second has a rerunnable exact certificate.

References

[1] J. W. Helton, I. Klep, S. McCullough and J. Volčič, *Noncommutative polynomials describing convex sets*, arXiv:1808.06669; J. Funct. Anal. 279 (2020), 108324.

[2] J. W. Helton, I. Klep and J. Volčič, *Geometry of free loci and factorization of noncommutative polynomials*, Adv. Math. 331 (2018), 589–626; arXiv:1708.05378.

[3] I. Klep and J. Volčič, *Free loci of matrix pencils and domains of noncommutative rational functions*, Comment. Math. Helv. 92 (2017), 105–130.